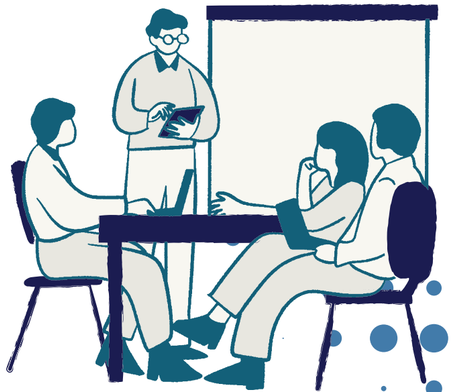
Valuez



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Why We Started *Valuez?*

- Property prices in Singapore depend on many factors (location, size, tenure, amenities, market conditions).
- Current property websites only show listings and past transactions, not future price insights.
- New buyers and investors often struggle to interpret data, creating a gap between information and actionable insights.

Build an interactive website that predicts property prices in Singapore.
Provide a tool that helps users make fact-based decisions when buying or selling properties.

Our Team

Zaf

Group Member

Adi

Group Member

Mandy

Group Leader

Noelle

Group Member

Angel

Group Member

Agenda Overview

1. Mission, Vision, and Product Objectives
2. Market Analysis
3. Research Summary
4. Overall Product Hierarchy
5. Development Tools and Methodology
6. Team Responsibilities

Our Mission and Vision

Our Mission

Our mission is to empower property buyers, sellers, and investors with reliable, data-driven insights through an intuitive real estate prediction website. By combining machine learning with geospatial and market data, we aim to bridge the gap between raw property information and actionable knowledge, making informed decision-making accessible to everyone.

Our Vision

Our vision is to build a user-friendly website that provides accurate property price predictions in Singapore. By combining machine learning with maps, trends, and nearby amenities, we aim to make property insights clear, accessible, and useful for buyers, sellers, and investors alike.

Product Objectives

- Collect & clean real estate data from reliable sources.
- Develop machine learning models for price prediction.
- Build a responsive, simple-to-use website interface.
- Integrate geospatial data (schools, MRT, amenities).
- Visualize data using maps for easy insights.
- Ensure smooth performance across devices.

Market Analysis

- **PropertyGuru:** Valuation + agent matching, but residential-focused.
- **SRX X-Value:** AI-powered, accurate for HDB/private homes, no industrial support.
- **Edge Fair Value:** Covers all property types, but commercial/industrial tools unreliable.
- **Propseller, HomerAI, Bluenest:** Provide quick valuations, but mainly residential with limited user control.

Market Analysis –
Basic Functions

Feature	PropertyGuru	SRX X-Value	EdgeProp (EFV)	Propseller Home Insights	Ohmyhome (HomerAI)	Bluenest MyHomeValue	Our product	
Basic Functions								
Free to use	✓	✓	✓	✓	✓	✓	✓	must have
Real-time price predicton	✓	✓	✓	☐	✓	☐	✓	must have
Machine Learning based	✓	✓	✓	✓	✓	✓	✓	must have
Basic property details (e.g., floor,size)	✓	✓	✓	✓	✓	✓	✓	must have
Geospatial insights (e.g., map, amenities, MRT)	☐	☐	✓	✓	☐	☐	✓	must have
Historical price trends	✓	☐	☐	☐	✓	☐	✓	must have
Data visualisation & user-friendly UI	✓	✓	✓	✓	✓	✓	✓	must have

Market Analysis – Enhanced Functions

Enhanced Functions								
Additional property detail (e.g., reno)	☐	☑	☐	☐	☐	☑	☑	nice to have
Rental estimate included	☑	☑	☑	☐	☑	☐	☑	nice to have
Comparable sales display	☐	☑	☑	☑	☑	☑	☑	must have
Registration/login required	☑	☑	☑	☑	☑	☑	☑	must have
Required the filtering of property type	☐	☑	☑	☑	☐	☐	☑	must have
Mobile responsive design	☐	☐	☑	☐	☑	☐	☑	must have

Market Analysis –
Advanced User Features

Advanced User Features								
Allow saving address	✓	✓	✓	☐	☐	☐	✓	must have
Allow saving searches	☐	☐	✓	✓	☐	☐	✓	must have
Allow saving property comparison	☐	✓	✓	☐	✓	☐	✓	must have
Supports agent access	✓	✓	✓	✓	✓	✓	✓	must have
Agent contact/integration	☐	✓	✓	✓	✓	✓	✓	must have
Financial calculators (stamp duty, loan, etc.)	✓	✓	✓	☐	✓	☐	✓	nice to have
Referral reward	✓	☐	✓	☐	✓	✓	✓	nice to have

Market Analysis –
Types of Valuation

Types of Valuation								
Commercial Valuation	☐	☐	☐	☐	☐	☐	☑	must have
Industrial Valuation	☐	☐	☐	☐	☐	☐	☑	must have

Research Summary

Market Gap Identified

There is a lack of real-time, open-access valuation tools for industrial and commercial properties such as warehouses, offices, and retail spaces.

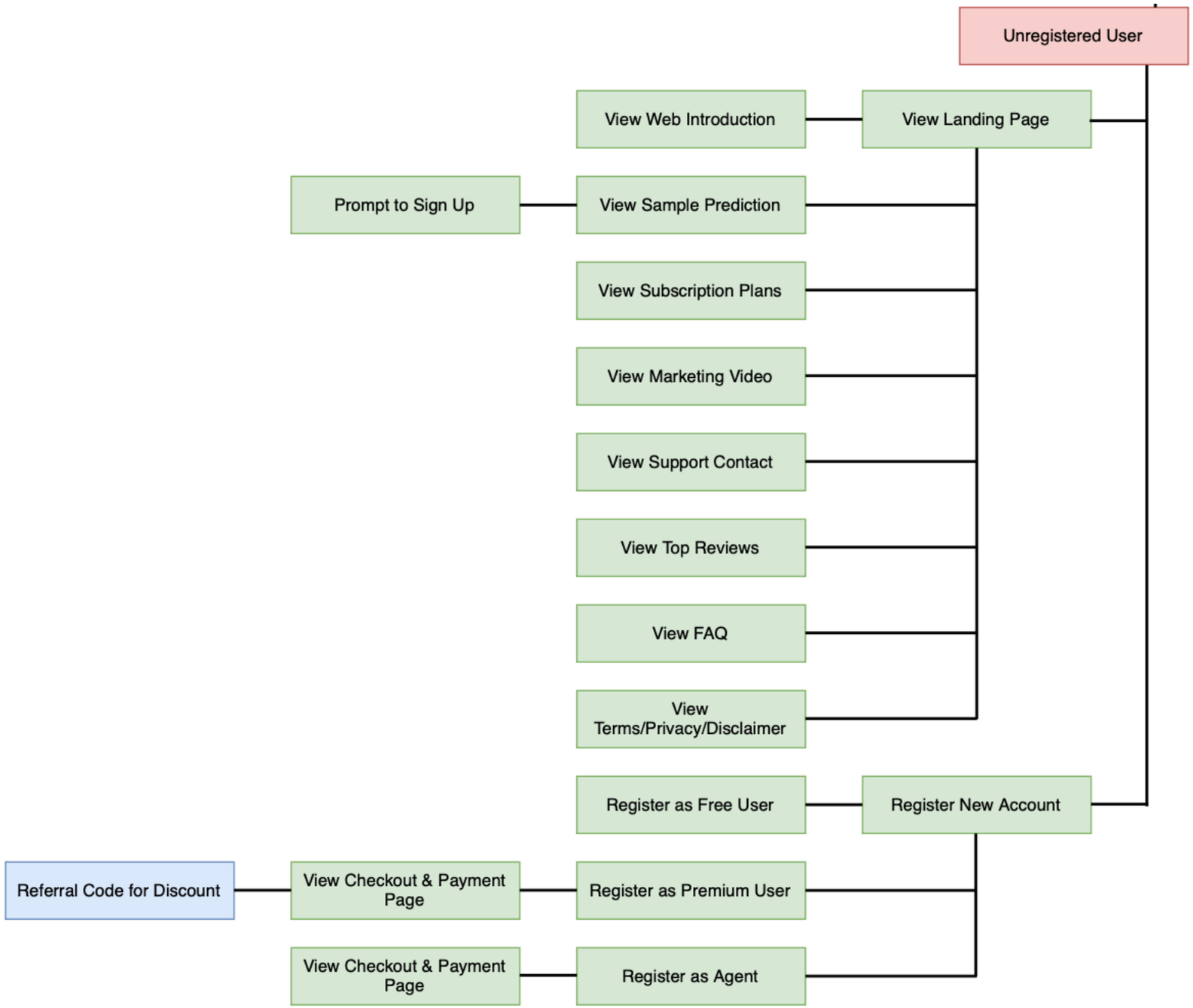
Our Platform's Differentiation

Instant predictions without agent callbacks or delays.
Supports industrial and commercial properties—an underserved market.
Machine learning-powered valuations based on URA's verified open data.
User-friendly reporting & comparison tools for deeper insights.

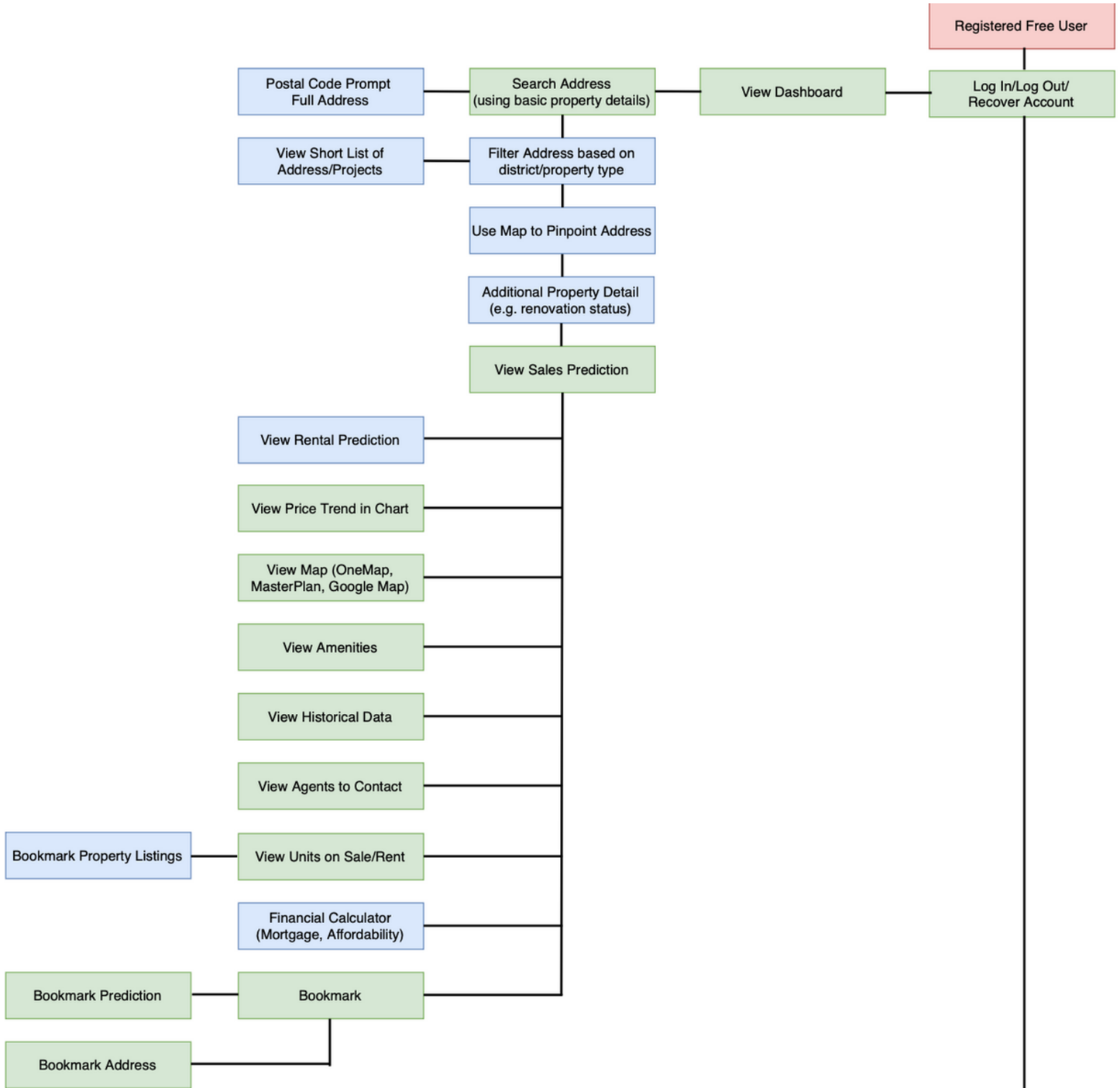
Unique Selling Point (USP)

Unlike most competitors, our system delivers fast, transparent, and reliable valuations for commercial and industrial properties, bridging a major gap in Singapore's PropTech market.

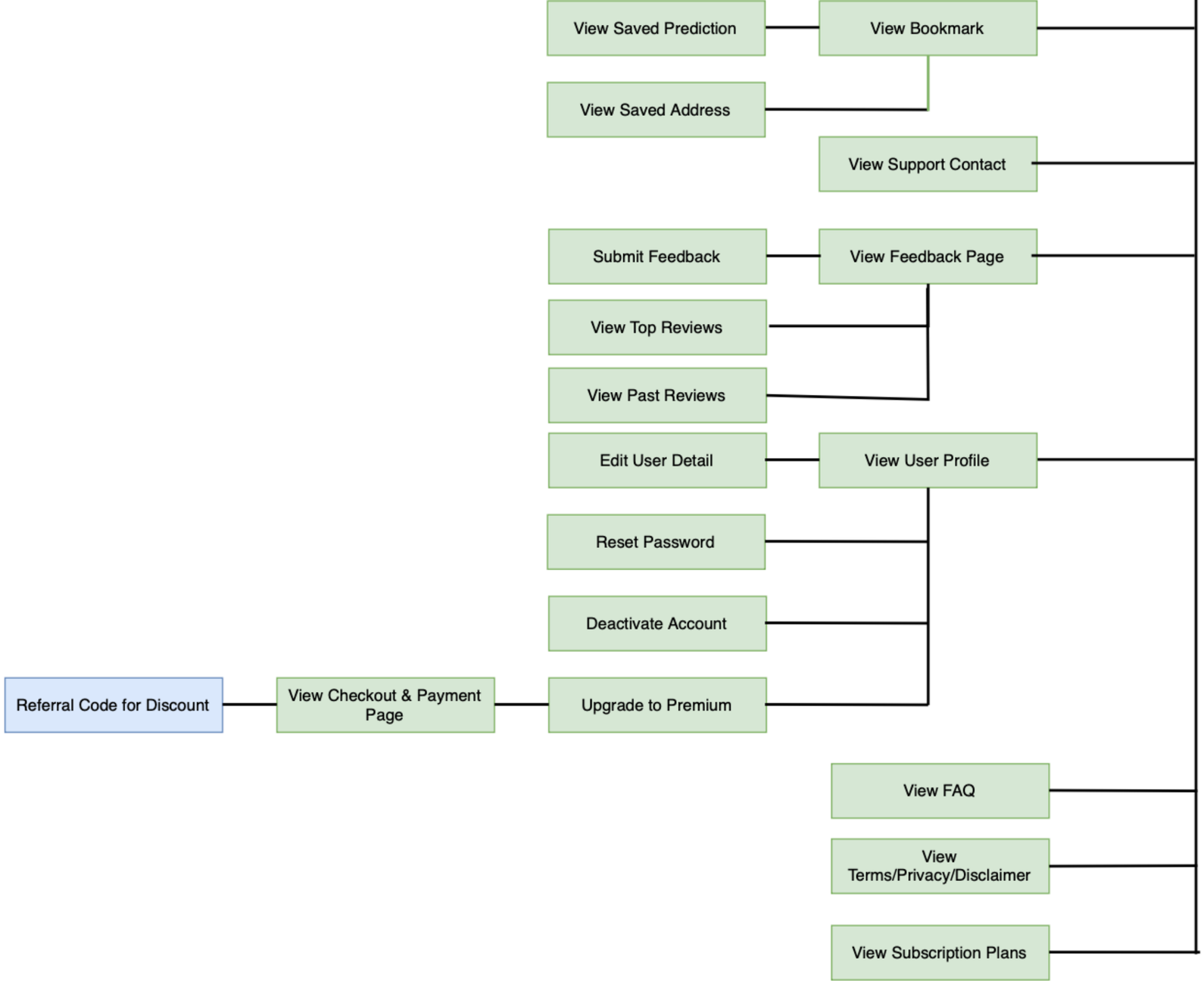
Overall Product Hierarchy – Unregistered User



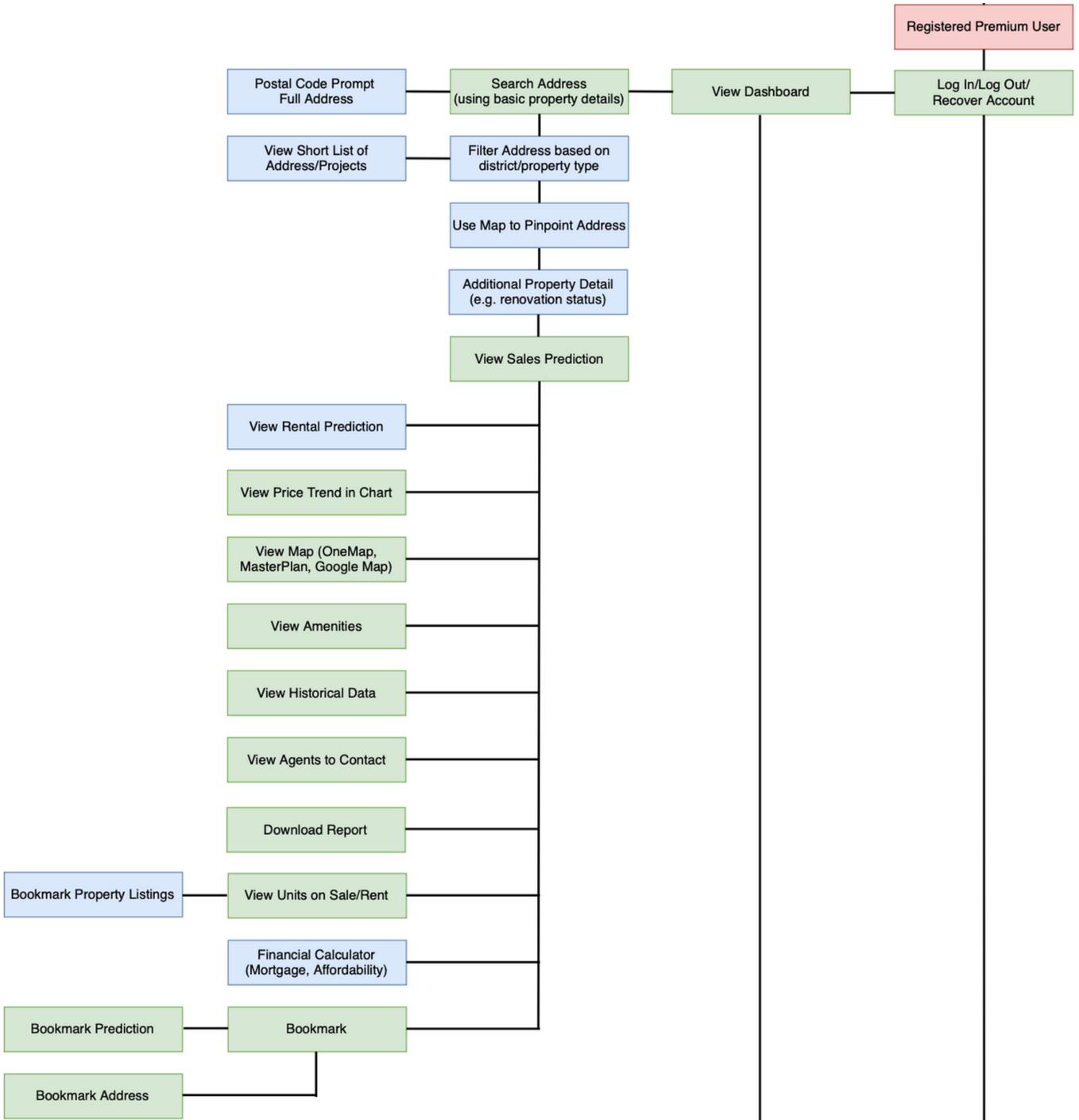
Overall Product Hierarchy - Registered Free User



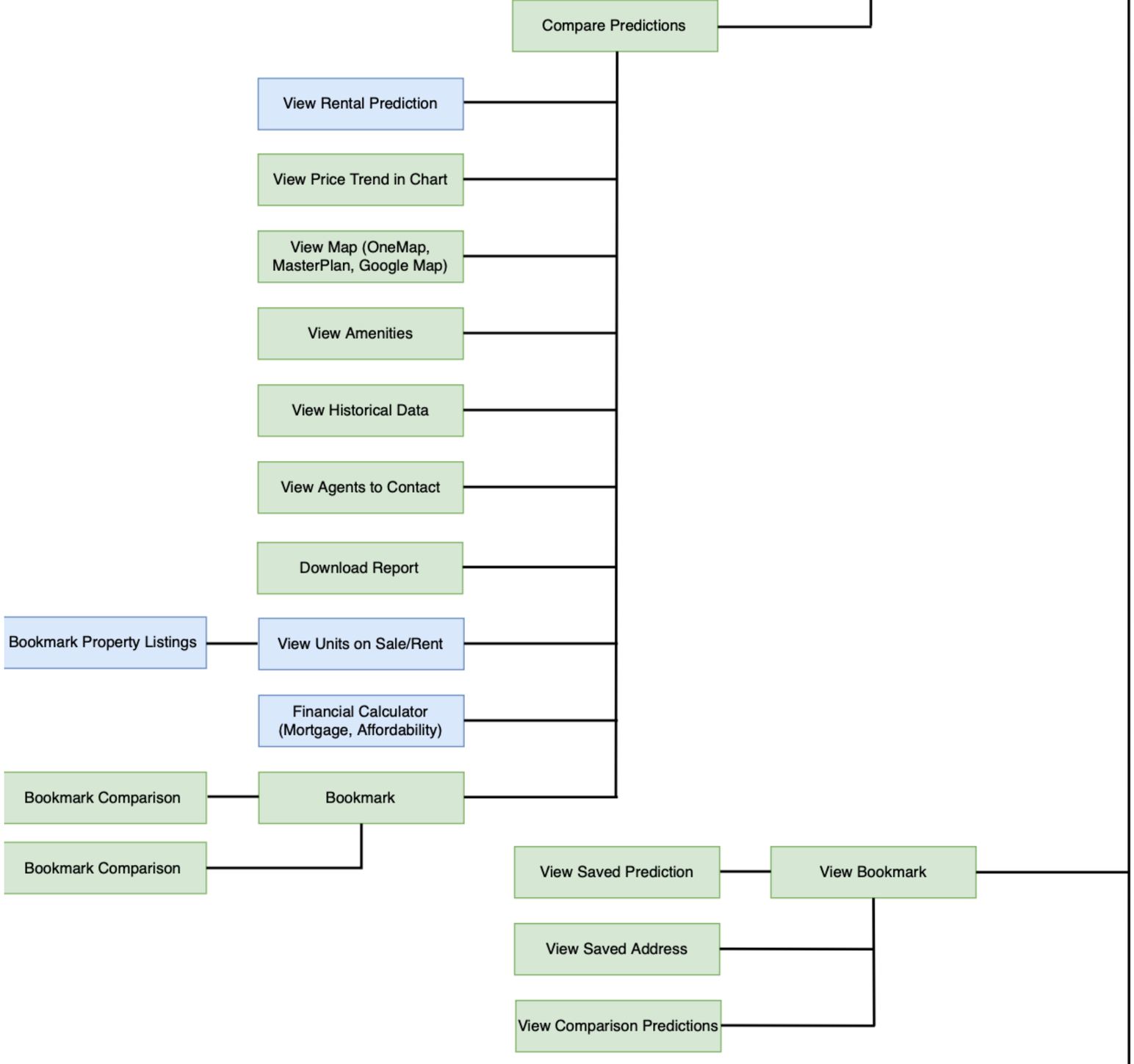
Overall Product Hierarchy - Registered Free User



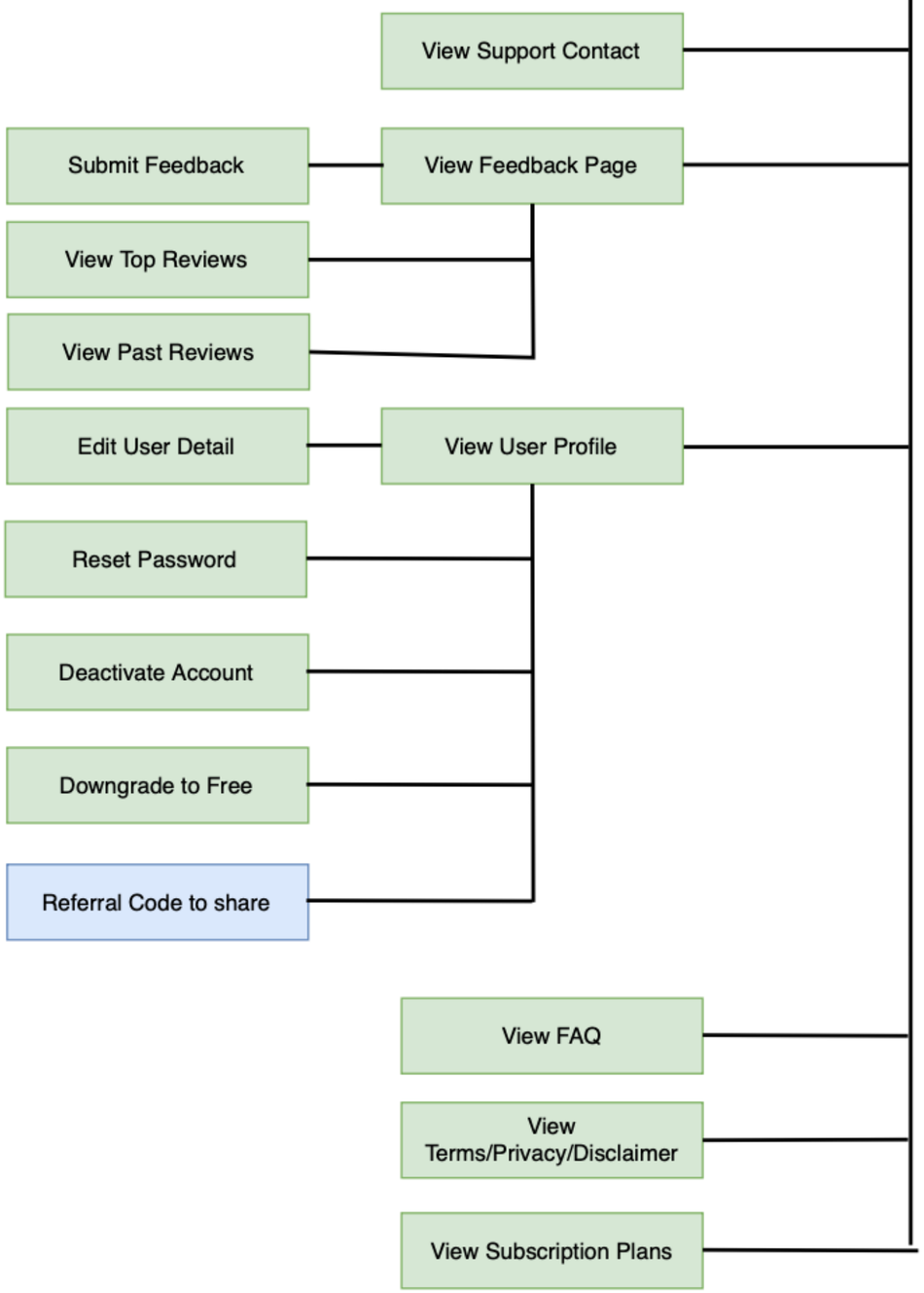
Overall Product Hierarchy - Registered Premium User



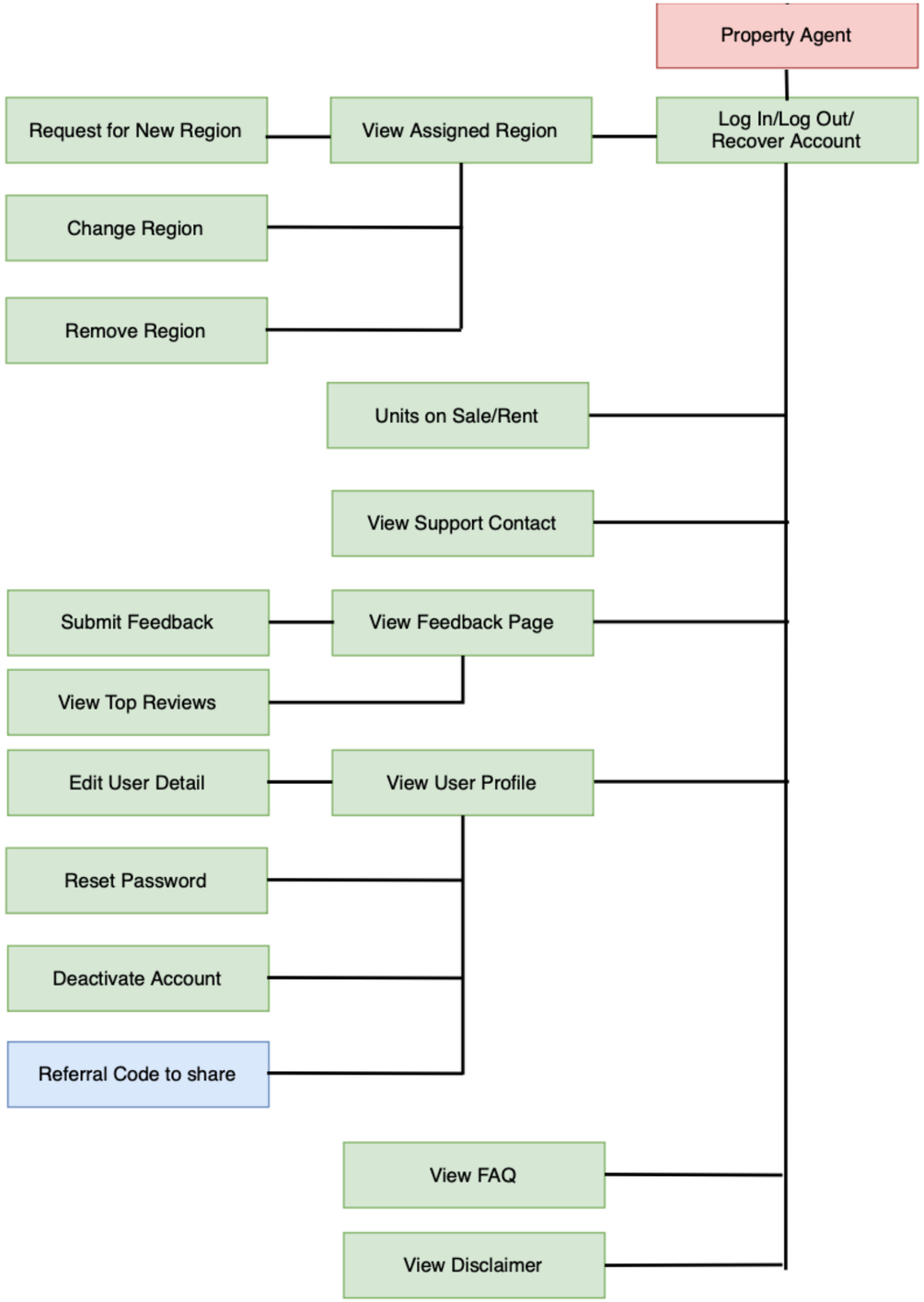
Overall Product Hierarchy - Registered Premium User



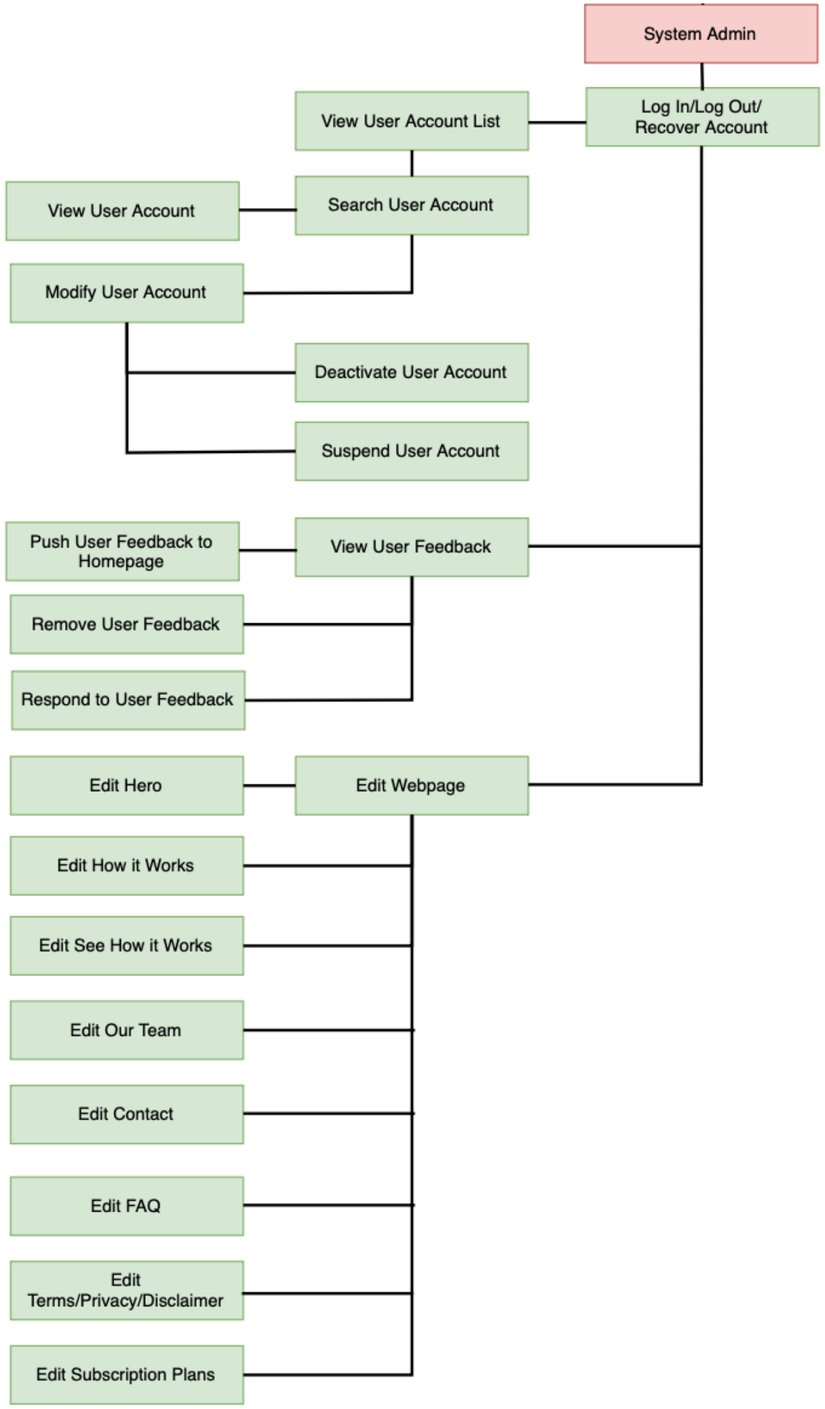
Overall Product Hierarchy - Registered Premium User



Overall Product Hierarchy – Property Agent



Overall Product Hierarchy – System Admin



Development Tools

Frontend:
React.js

Backend:
Flask (Python)

Database:
PostgreSQL + PostGIS (for geospatial queries)

APIs: Google Maps, URA Map API

Machine Learning Libraries: Scikit-learn, XGBoost, SHAP, GeoPandas, NumPy

Hosting: Backend & Database on Render, Frontend on Vercel

Server:
Render Web Service + Database Instance

Deployment: GitHub CI/CD (auto-deploy to Render & Vercel)

Development Methodology

Scrum + Taiga were chosen to manage our project efficiently:

Adaptability & Iteration

- Work in short sprints to refine predictions & add new features.
- Sprint reviews/retrospectives in Taiga help adjust priorities quickly.

Task Visualization

- Taiga's Kanban board & backlog track progress (data, model, UI, API).
- Ensures all components are delivered on time.

Streamlined Feedback

- Sprint reviews ensure early detection of issues (data, system, UI).
- Continuous improvement through stakeholder feedback.

User Stories Integration

- Stories guide development through Taiga-linked tasks.

Customizable Workflows

- Tailored stages: data gathering → model building → integration → testing.
- Easy to adjust for new features (e.g., premium insights, filters).

Remote Collaboration

Taiga tools (comments, attachments, notifications) keep the team aligned.

Focus on Impact

Scrum prioritizes tasks that deliver most value (accuracy, usability, visuals).

Team Responsibilities (RACI Matrix)

Activities/Deliverable	Project Leader/Front-End Developer (Mandy Ng Xue Ling)	Documentation/ Front-End Developer (Thet Myat Noe)	Documentation/ Front-End Developer (Angel Lim)	Documentation/ Back-End Developer (Zafeerah Akbar)	Documentation/ Back-End Developer (Aung Myint Myat)
Technical Design Manual	R/C	R	R	R	R
Website Frontend/Backend	R/C	R	R	R	R
Project Requirement Document	R/C	R	R	R	R
User Requirement Specification	R/C	R	R	R	R
Preliminary Tech Documentation	R/C	R	R	R	R

R = Responsible

A = Accountable

C = Consulted

I = Informed

Prototype Demo